

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think>

Cost to plant + annual cost * years = income * years

$$90 + 3y = 1.5 * 7y$$

$$90 = 3y(1.5 - 3)$$

$$90 / (-1.5 * 3 + 3) = 90 / -1.5 = 6 \text{ years.}$$

<answer> 6 </answer>

